GROUP NO- 07
Div : D20A
Roll No : 39, 48, 51, 57

IOE EXPERIMENT 07

Aim: Publish and Subscribe data using MQTT protocol.

COMPONENTS:

Hardware: None
Software: Eclipse Mosquitto (Broker), Command Prompt (CLI)

THEORY:

MQTT (Message Queuing Telemetry Transport) is a lightweight and efficient communication protocol designed for devices that have limited power and processing capabilities. It is widely used in IoT (Internet of Things) applications due to its low bandwidth usage and simple structure. MQTT uses the publish-subscribe model for exchanging data between devices.

In a publish-subscribe model, communication takes place through a central component called a broker. Devices (called clients) do not communicate directly with each other. Instead:

- A publisher sends messages to a topic on the broker.
- A subscriber receives messages from the broker by subscribing to that same topic.

This approach is highly scalable and decouples the sender and receiver, which makes it suitable for networks where devices frequently connect and disconnect, such as IoT systems.

Eclipse Mosquitto is one of the most popular open-source MQTT brokers. It implements versions 3.1, 3.1.1, and 5.0 of the MQTT protocol. Mosquitto is lightweight and can run on devices ranging from microcontrollers to full servers. It comes with command-line tools like:

- `mosquitto`: for starting the broker
- `mosquitto_pub`: to act as a publisher
- `mosquitto_sub`: to act as a subscriber

Using Mosquitto, multiple clients can publish and subscribe to various topics, and the broker efficiently handles message routing. This makes it ideal for sensor data collection, smart home automation, and real-time monitoring systems.

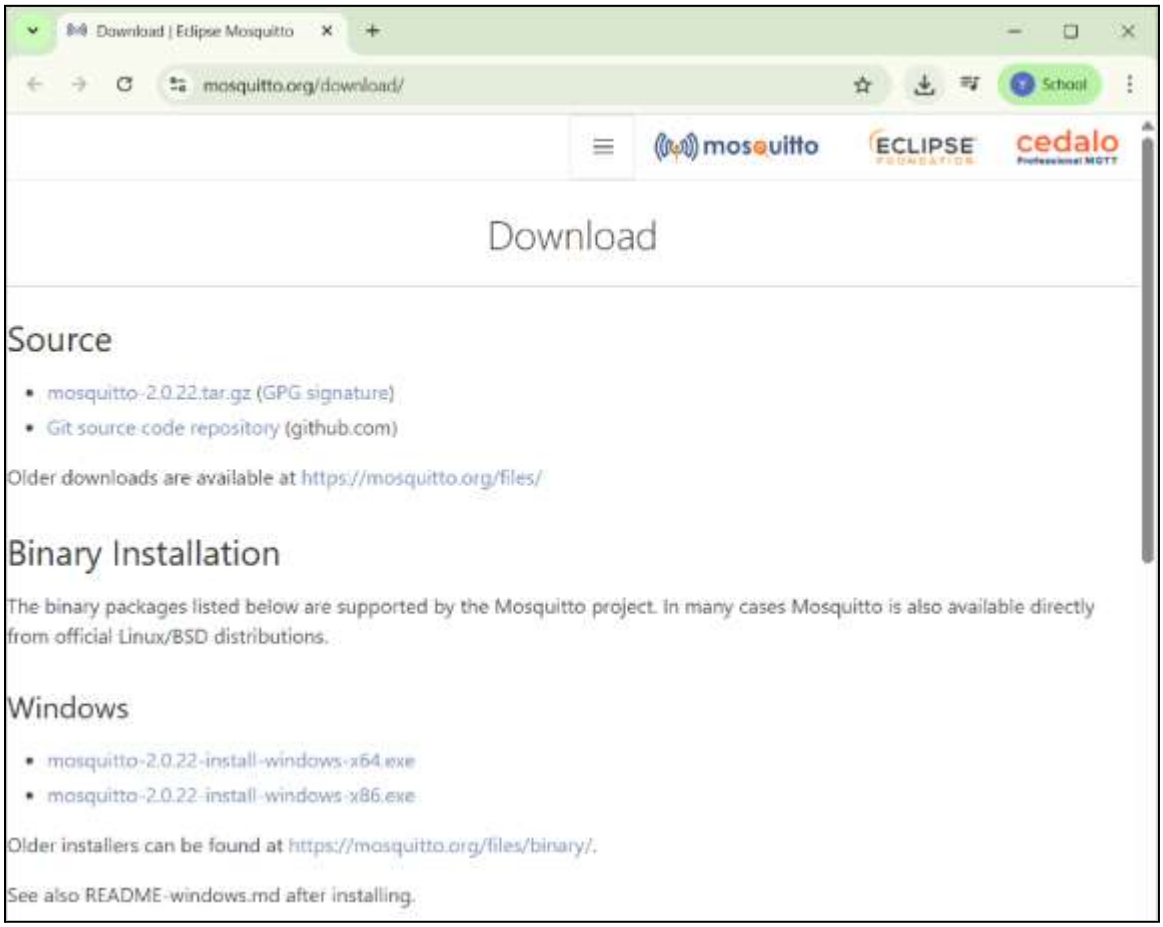
MQTT operates over TCP/IP and typically uses port 1883. It ensures reliable delivery using different levels of Quality of Service (QoS), depending on the need for message confirmation or retransmission.

Overall, MQTT with Mosquitto is a powerful combination for learning how IoT devices communicate, especially in situations where resources are limited and reliable communication is essential.

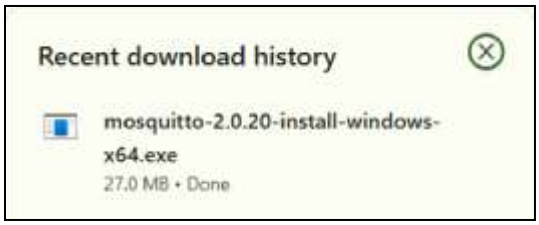
STEPS:

Step 1: Download Mosquitto Broker for Windows

1. Open your browser and go to the official Mosquitto download page: https://mosquitto.org/download



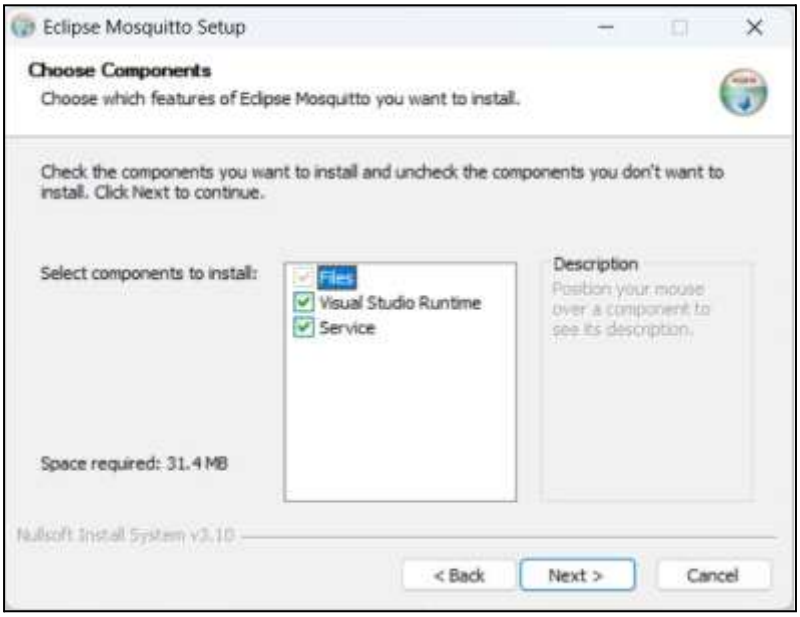
2. Scroll down to the Windows section and download the .exe installer for Windows.

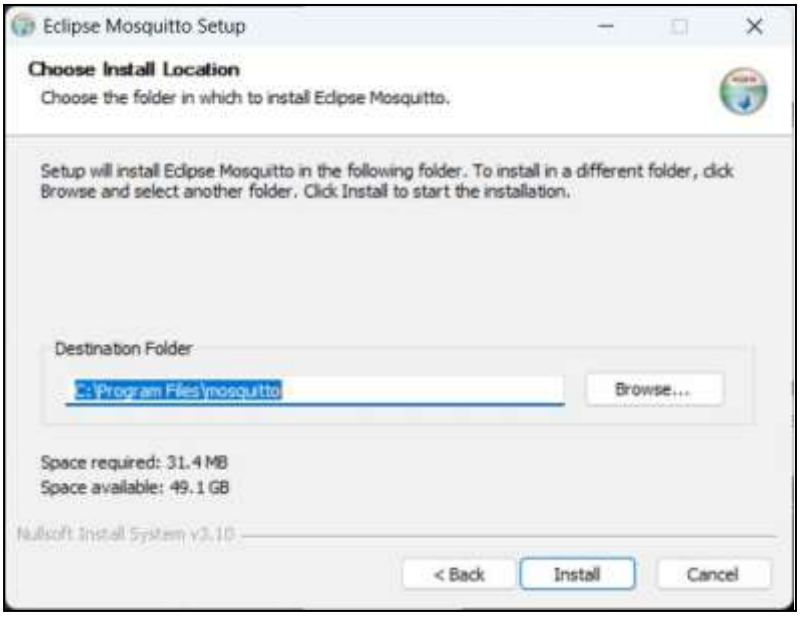


Step 2: Install Mosquitto Broker

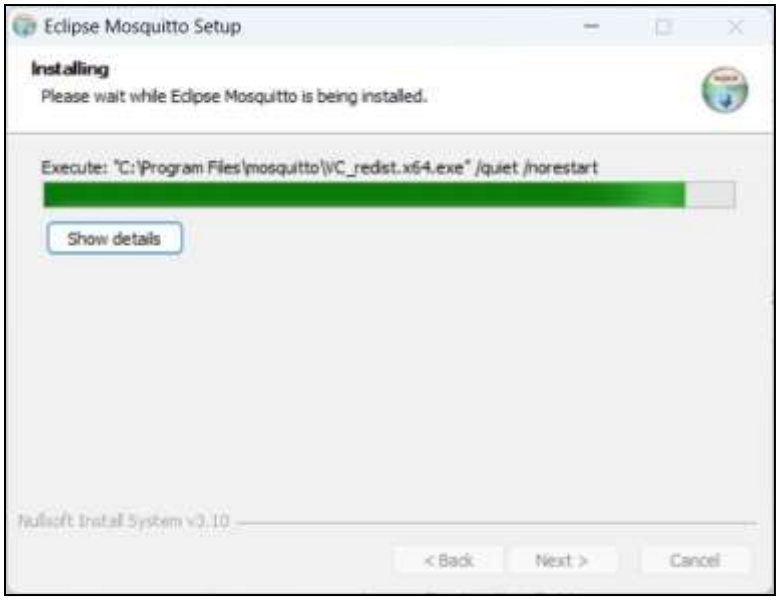
1. Run the downloaded installer.

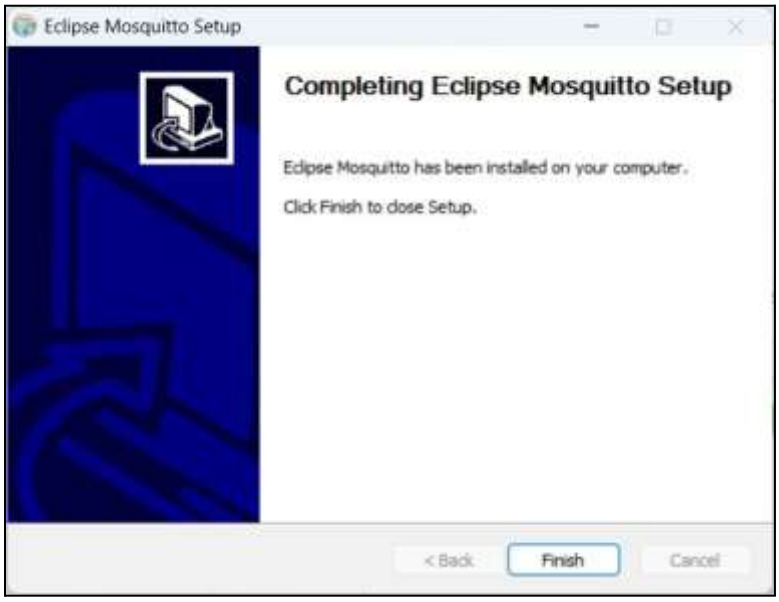
2. During installation:





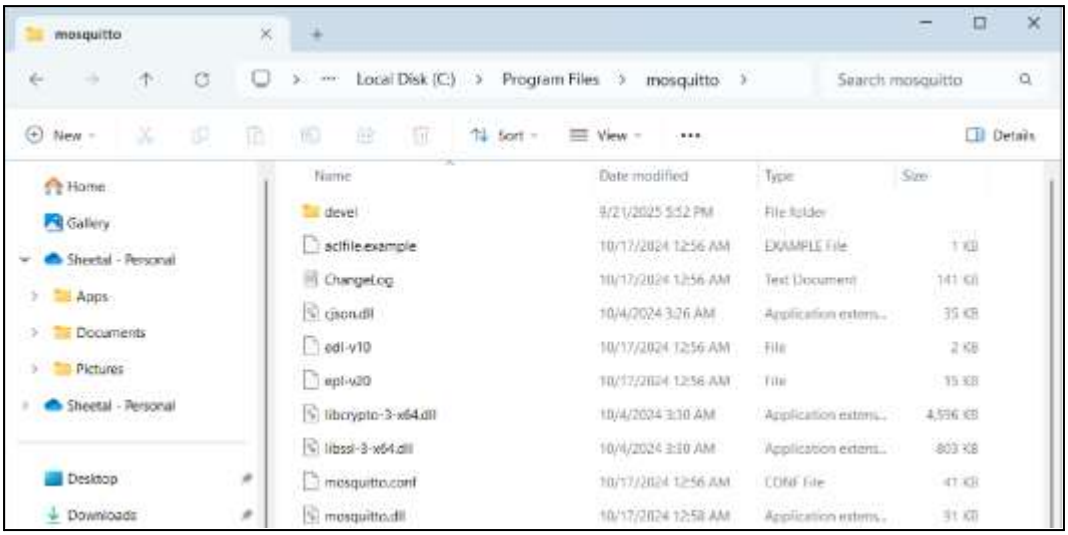
3. Complete the installation.



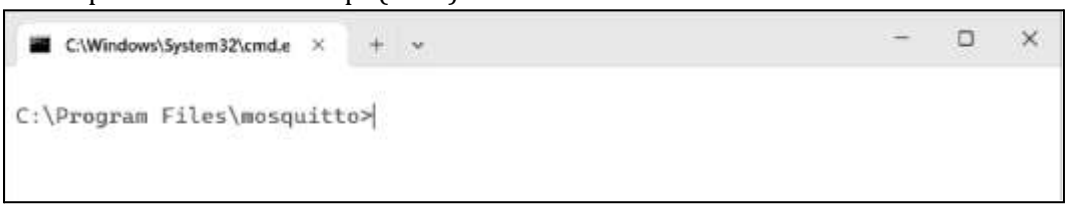


Step 3: Open Command Prompt in Mosquitto Directory

1. Navigate to the Mosquitto installation directory



2. Open Command Prompt (CMD).



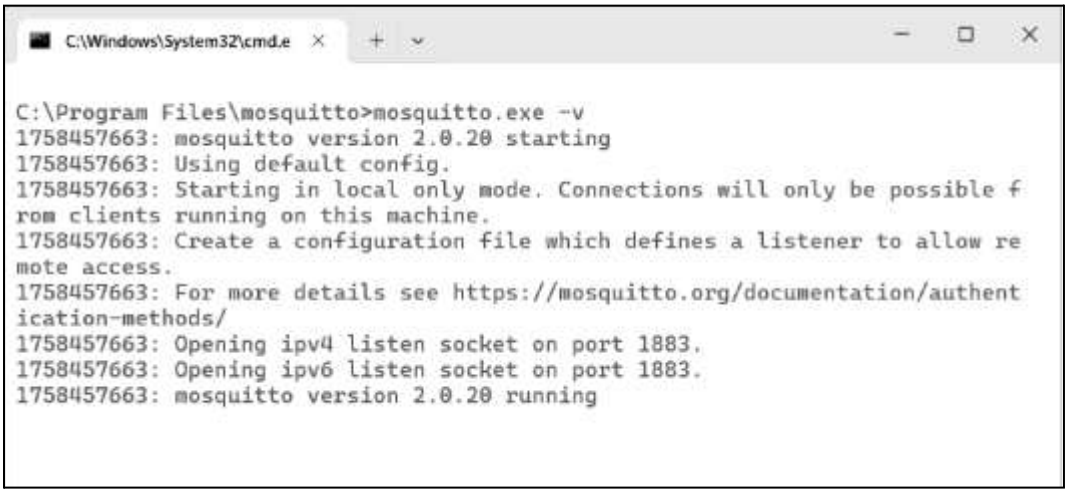
Step 4: Start the Mosquitto Broker

1. Run the broker using the following command:

```
mosquitto.exe -v
```

- -v stands for verbose mode (shows logs and connections). •

The broker listens on port 1883 by default.



Step 5: Open a New CMD – Start the Subscriber 1.

Navigate to Mosquitto folder:

2. Open a new Command Prompt window.
3. To subscribe to a topic (basic usage):

mosquitto_sub.exe -t testTopic

```
C:\Windows\System32\cmd.e

C:\Program Files\mosquitto>mosquitto_sub.exe -t gavy
```

4. To subscribe to a broker running on a different IP and port, use:

mosquitto_sub.exe -h 140.2.66.130 -p 1234 -t testTopic

```
C:\Windows\System32\cmd.e

C:\Program Files\mosquitto>mosquitto_sub.exe -t gavy -h 140.2.66.130 -p 1234
```

- -h: Host IP of the MQTT broker (here it's 140.2.66.130)
- -p: Port number used by the broker (here it's 1234)
- -t: Topic name

Step 6: Open Another CMD – Start the Publisher 1.

Navigate to the Mosquitto folder again

2. Open another Command Prompt window (3rd window).
3. Publish a message using:

mosquitto_pub.exe -t testTopic -m "Hello from Publisher"

- -t is the topic name (same as subscriber).
- -m is the message content.

```
C:\Windows\System32\cmd.e

C:\Program Files\mosquitto>mosquitto_pub.exe -t gavy -m "19_39_41_49"

C:\Program Files\mosquitto>
```

5. To publish with debug output:

mosquitto_pub.exe -t testTopic -m "Hello with debug" -d

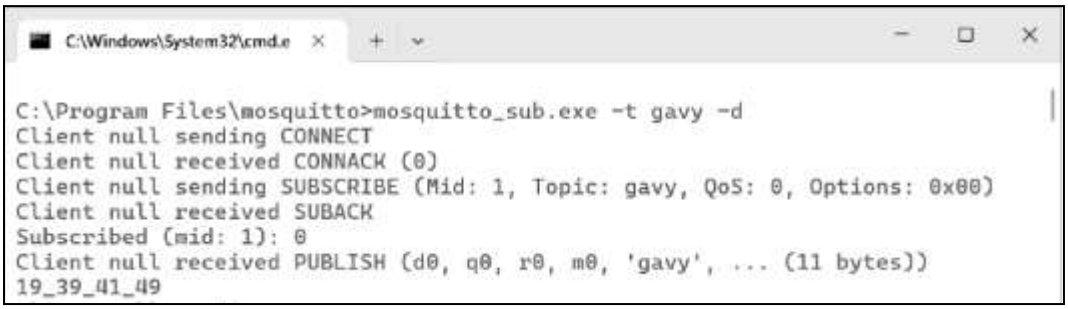
```
C:\Windows\System32\cmd.e

C:\Program Files\mosquitto>mosquitto_pub.exe -t gavy -m "19_39_41_49" -d
Client null sending CONNECT
Client null received CONNACK (0)
Client null sending PUBLISH (d0, q0, r0, m1, 'gavy', ... (11 bytes))
Client null sending DISCONNECT

C:\Program Files\mosquitto>
```

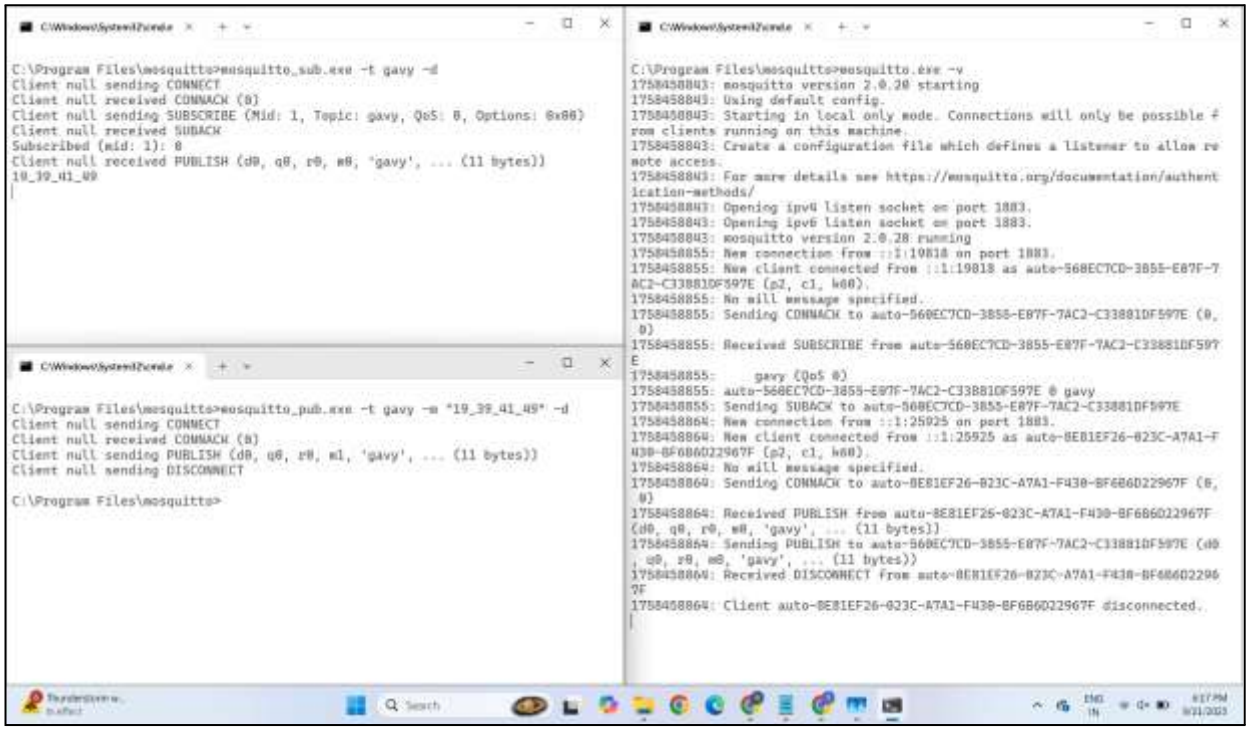
Step 7: Verify Output

- Now check the subscriber CMD window.
- You should see the message "Hello from Publisher" displayed.

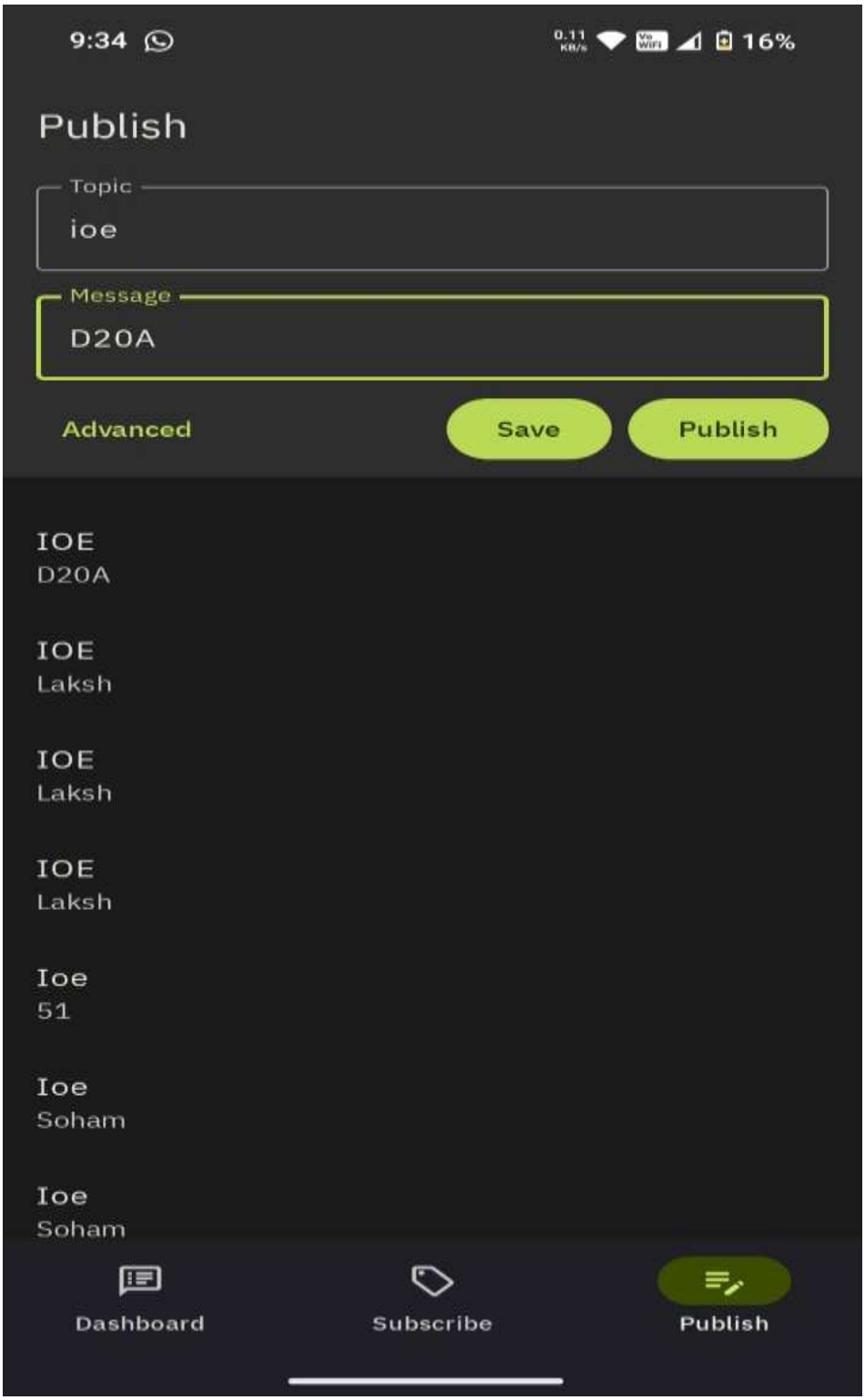


Step 8: All Windows Together

Screenshot showing all three CMD windows (broker, publisher, and subscriber) running at the same time. This helps show the full working setup.



Step 9: In mobile application: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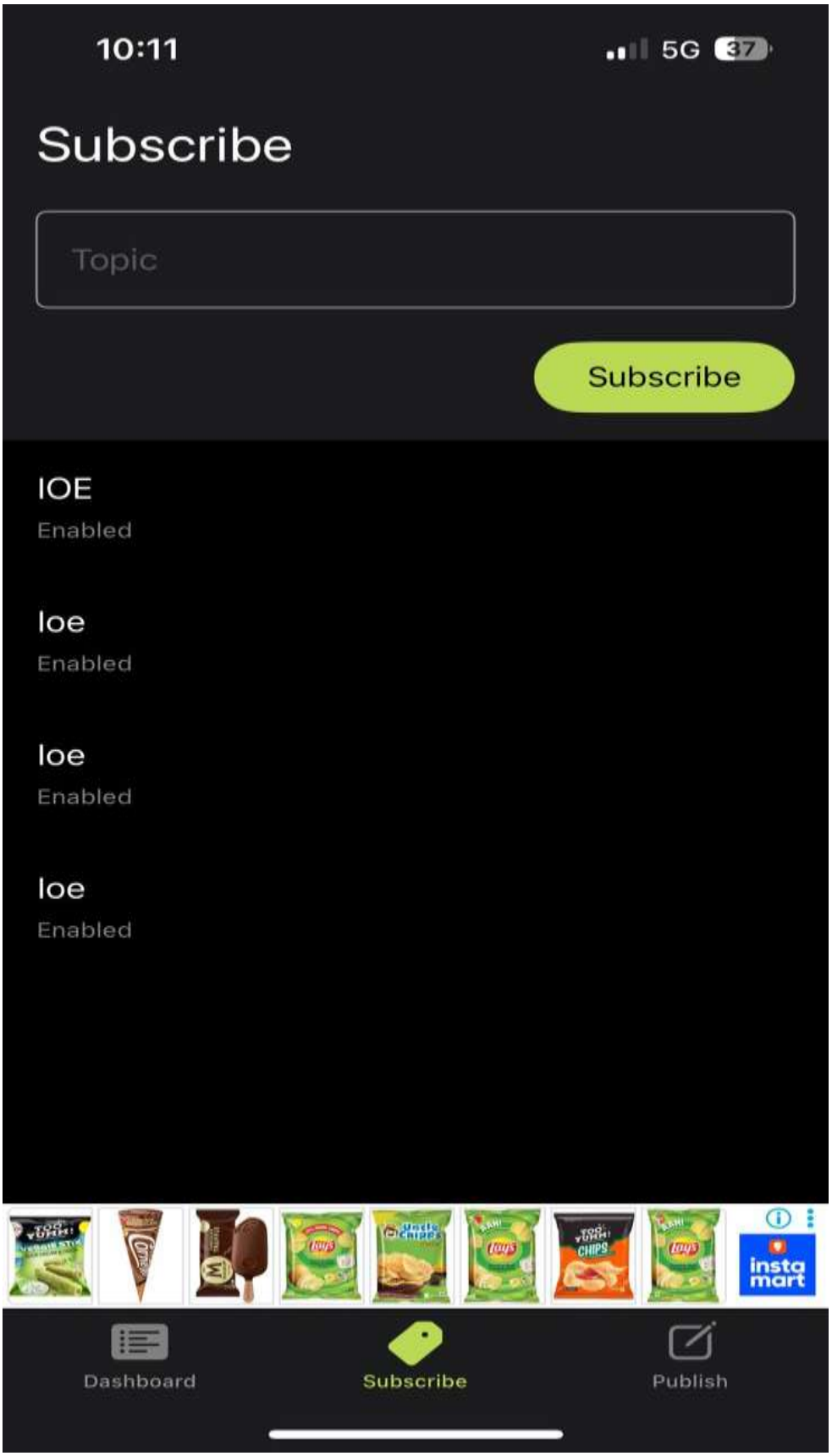
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OBSERVATION:

- The subscriber successfully received messages sent by the publisher via the broker.
- All components (broker, publisher, subscriber) ran correctly on the same machine.
- Communication was established using the MQTT protocol over port 1883.

CONCLUSION:

In this experiment, we learned how to set up an MQTT-based publish-subscribe system using the Eclipse Mosquitto broker. It helped us understand the basics of MQTT communication and how different components (publisher, subscriber, and broker) interact with each other in an IoT system.